

Development of prior distributions for sablefish catchability

Dana Hanselman

Introduction

In the Alaska sablefish stock assessment, prior distributions are used to apply knowledge from outside of the model to assist in determining parameters that are difficult to estimate. Catchability (q) is one of these key parameters that has large consequences to the model as it is directly related to resultant biomass estimates. Currently, the prior distributions in the model are based on previous model estimates of catchability with an imprecise distribution (CV=500%). In this analysis, we use NMFS trawl survey biomass estimates to estimate longline survey and fishery catchability and to estimate the relative catchability of the GOA trawl survey (<500 meters in depth) to total trawl estimated biomass. These values can then be translated into the model as prior distributions for estimating catchability of each abundance index.

Methods

NMFS has bottom trawl sablefish biomass estimates for the Gulf of Alaska and the Aleutian Islands since 1980 on a triennial, then biennial basis. We use a combination of these two surveys biomass estimates, in addition to an average adjustment for biomass contained on the Bering Sea slope as an estimate of true biomass. This adjustment is used in lieu of a consistent trawl survey of the Bering Sea slope. We then use the ratio of each abundance index in the model to these estimates as an estimate of catchability. The formula for each yearly catchability index is:

$$\hat{q}_{iy} = \frac{\hat{B}_{iy}}{\hat{B}_{trawl,y} (1 + p(BS))}$$

where $\hat{q}_{iy}$ is the estimate of catchability for the i th index in year y , $\hat{B}_{iy}$ is the estimated biomass for the i th index in year y , $\hat{B}_{trawl,y} (1 + p(BS))$, is the estimated trawl biomass for Gulf of Alaska, Aleutian Islands, adjusted upward by the average proportion of biomass in the Bering Sea, $p(BS)$, estimated by the longline survey.

The variance of the annual catchability is estimated with the delta method (Shou 2002).

$$\text{var}(\hat{q}_{iy}) = \frac{\hat{B}_{iy}^2}{(1 + p(BS))^2 \hat{B}_{trawl,y}^2} \left[\frac{\text{var}(\hat{B}_{iy})}{\hat{B}_{iy}^2} + \frac{(1 + p(BS))^2 \text{var}(\hat{B}_{trawl,y})}{(1 + p(BS))^2 \hat{B}_{trawl,y}^2} \right]$$

The formula for the overall catchability prior mean is:

$$\hat{q}_i = \frac{\sum \hat{q}_{iy}}{n}$$

and then the coefficient of variation for the prior mean is found from the variance of two stage sampling (Cochran 1977), ignoring the finite population correction is approximated by:

$$CV(\hat{q}_i) = \frac{\sqrt{\text{var}(\hat{q}_i)/n + \text{var}(\hat{q}_{iy})/m}}{\hat{q}}$$

where $\text{var}(\hat{q}_i)$ is the variance among means, and $\text{var}(\hat{q}_{iy})$ is the variance within means shown above.

These results are then used as the prior mean and coefficient of variation for lognormal prior distributions to be applied in the sablefish stock assessment as an alternative to the current diffuse prior distributions being used.

Results

Catchability estimates were computed for each abundance index used in the stock assessment model (Table 1). Lognormal priors were constructed with the means and CVs derived from the analysis. These distributions are shown in Figure 1. The catchability mean for the domestic longline survey is higher than the Japanese longline survey index and domestic fishery index which corroborates previous estimates in Kimura and Zenger (1997). The mean value for the NMFS GOA trawl survey index is higher (0.7) than the mean value previously used for the diffuse prior used in the 2007 sablefish assessment (0.3).

Discussion

New prior distributions for catchability appear to be reasonable values, and are within range of previous values used in the sablefish stock assessment. The most important assumption in this analysis that may be violated is that the trawl survey catchability is equal to one. It is more likely that the true catchability of the trawl survey is less than, rather than more than one, both because sablefish are fast swimmers and because the trawl survey has limited coverage of the full depth distribution of sablefish. If this is true, then assuming a trawl catchability of one is a precautionary assumption. However, there is enough uncertainty in the derived prior distributions for the data to provide substantial influence on the final estimate, yet there is enough precision to guide the model on how each catchability value is related between indices.

Applying these distributions to the model should result in greater model stability, and more precise estimates of biomass. The effect on harvest recommendations will likely be small, but directionality is not obvious due to interaction among the catchabilities and between other parameters such as selectivity.

References

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Table 1. Values and results used in catchability estimation for each index.

NMFS Domestic LL survey								
Year	$\hat{B}_{iy}^2$	$\sqrt{\text{var}(\hat{B}_{iy})}$	$\hat{B}_{trawl,y}$	$\hat{q}_{iy}$	$\text{var}(\hat{q}_{iy})$	$\sqrt{\text{var}(\hat{q}_{iy})}$	CV	
1990	2141	214.1	222.563	9.620	3.673	1.916	20%	
1993	1894	189.4	253.168	7.481	1.957	1.399	19%	
1996	2017	201.7	151.562	13.308	6.560	2.561	19%	
1999	1740	174	180.328	9.649	1.536	1.239	13%	
2001	1798	179.8	195.054	9.218	6.001	2.450	27%	
2003	1759	175.9	247.355	7.111	1.094	1.046	15%	
2005	1695	169.5	278.618	6.084	0.675	0.822	14%	

$(1 + p(BS))$	1.119		
$\hat{q}_{iy}$	8.924	Grand Variance	8.696
$\sqrt{\text{var}(\hat{q}_i)}$	2.372	Grand Stdev	2.949
CV	27%		
$\hat{q}_i$	7.979	Grand CV	33%

Table 1 (continued). Values and results used in catchability estimation for each index.

Japanese LL survey							
Year	$\hat{B}_{iy}^2$	$\sqrt{\text{var}(\hat{B}_{iy})}$	$\hat{B}_{\text{trawl},y}$	$\hat{q}_{iy}$	$\text{var}(\hat{q}_{iy})$	$\sqrt{\text{var}(\hat{q}_{iy})}$	CV
1984	1804	180.4	402.221	4.485	0.457	0.676	15%
1987	2068	206.8	405.620	5.098	0.576	0.759	15%
1990	1454	145.4	222.563	6.533	1.694	1.301	20%
1993	1318	131.8	253.168	5.206	0.947	0.973	19%
$(1 + p(BS))$			1.119				
$\hat{q}_{iy}$			5.331	Grand Variance	1.662		
$\sqrt{\text{var}(\hat{q}_i)}$			0.862	Grand Stdev	1.289		
CV			16%				
$\hat{q}_i$			4.766	Grand CV	24%		

NMFS Domestic Fishery CPUE							
Year	$\hat{B}_{iy}^2$	$\sqrt{\text{var}(\hat{B}_{iy})}$	$\hat{B}_{\text{trawl},y}$	$\hat{q}_{iy}$	$\text{var}(\hat{q}_{iy})$	$\sqrt{\text{var}(\hat{q}_{iy})}$	CV
1990	1201	120.1	222.563	5.396	1.156	1.075	20%
1993	904	90.4	253.168	3.571	0.446	0.668	19%
1996	1201	120.1	151.562	7.924	2.326	1.525	19%
1999	1316	131.5729	180.328	7.296	0.878	0.937	13%
2001	1110	111.0479	195.054	5.693	2.289	1.513	27%
2003	1218	121.765	247.355	4.923	0.524	0.724	15%
2005	1307	130.69	278.618	4.691	0.401	0.634	14%
$(1 + p(BS))$			1.119				
$\hat{q}_{iy}$			5.642	Grand Variance	3.433		
$\sqrt{\text{var}(\hat{q}_i)}$			1.512	Grand Stdev	1.853		
CV			27%				
$\hat{q}_i$			5.044	Grand CV	33%		

Table 1 (continued). Values and results used in catchability estimation for each index.

NMFS GOA Trawl survey (<500m)							
Year	$\hat{B}_{iy}^2$	$\sqrt{\text{var}(\hat{B}_{iy})}$	$\hat{B}_{\text{trawl},y}$	$\hat{q}_{iy}$	$\text{var}(\hat{q}_{iy})$	$\sqrt{\text{var}(\hat{q}_{iy})}$	CV
1984	294.429	43.53	402.221	0.732	0.019	0.136	19%
1987	271.099	38.17	405.620	0.668	0.014	0.120	18%
1990	213.882	37.57	222.563	0.961	0.056	0.236	25%
1993	249.516	39.98	253.168	0.986	0.049	0.222	23%
1996	144.808	24.83	151.562	0.955	0.052	0.227	24%
1999	103.766	10.68	180.328	0.575	0.006	0.075	13%
2003	189.184	25.67	247.355	0.765	0.018	0.133	17%
2005	178.884	23.25	278.618	0.642	0.010	0.102	16%
$(1 + p(BS))$			1.119				
$\hat{q}_{iy}$			0.786	Grand			
$\sqrt{\text{var}(\hat{q}_i)}$			0.161	Variance	0.054		
CV			20%	Grand			
$\hat{q}_i$			0.702	Stdev	0.232		
				CV	30%		

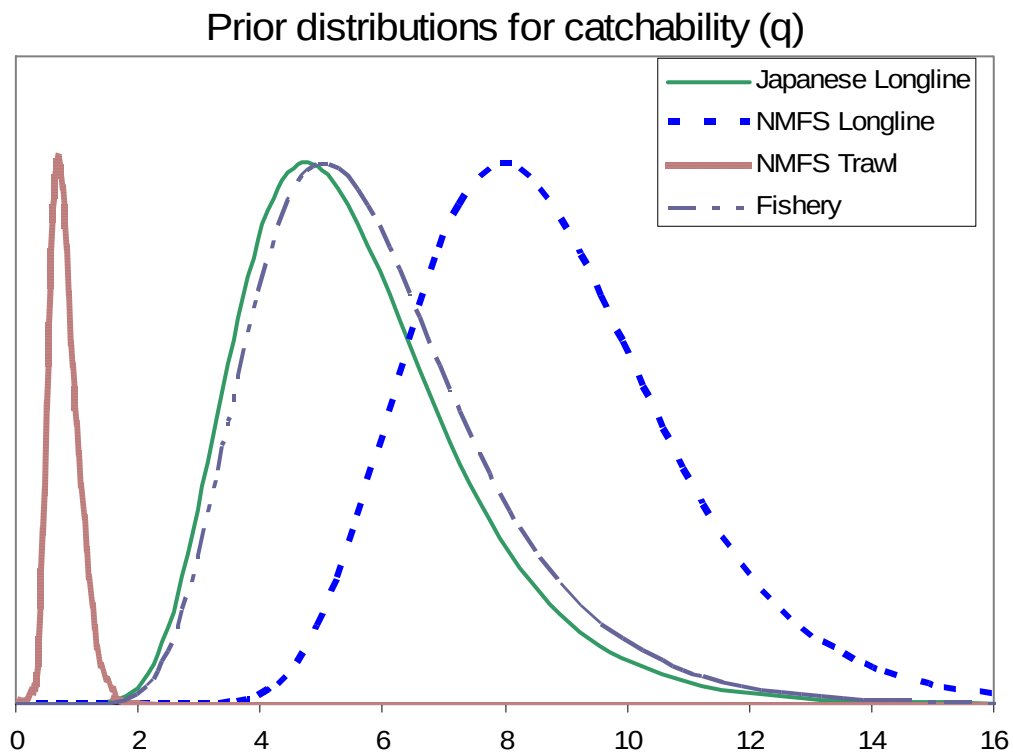


Figure 1. Prior distributions for catchability for four sablefish abundance indices.